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Resolving the OP3 Gate: A Numerical Campaign to Validate the Screening Coefficient Before Cryostat Fabrication

Methodological Framework: Evaluation of the MPO-TDVP Simulation Strategy

The central task of this research is to evaluate the feasibility of achieving a screening coefficient, $\alpha_{screen}(50, T_\phi) \geq 10^{-2}$, for a 50-qubit transmon array operating under cryogenic conditions. This evaluation serves as the critical "Go/No-Go" gate (OP3 Stage 3) for committing to the expensive fabrication of a cryostat [4]. The proposed methodology for this evaluation is a rigorous numerical simulation based on the Matrix Product Operator (MPO) formalism combined with the Time-Dependent Variational Principle (TDVP). This approach is designed to simulate the time evolution of the open quantum system, which consists of the transmon array coupled to an Ohmic bosonic bath at a temperature of $T = 15$ mK [7]. The success of this endeavor hinges on the proper implementation of several key components: the representation of the system Hamiltonian and its dissipative coupling, the choice of a suitable time-evolution algorithm, and the careful management of computational resources while preserving essential physical observables.

The total Hamiltonian of the system is composed of three parts: the system Hamiltonian (H_{sys}), which describes the 50 coupled transmon qubits; the bath Hamiltonian (H_{bath}), representing the surrounding electromagnetic environment as a collection of harmonic oscillators; and the interaction Hamiltonian (H_{int}), which models the coupling between the qubits and the bath [42] [71]. For superconducting qubits like the transmon, the environment is often modeled as an Ohmic bath, characterized by a spectral density function $\mathcal{J}(\omega) \propto \omega$ [7] [8]. The specific parameters provided are an Ohmic spectrum with a cutoff frequency $\omega_c = 5$ GHz, which sets the characteristic energy scale of the environment, and a temperature of $T = 15$ mK, chosen to suppress thermal excitations and ensure the system starts in its ground state or a well-defined low-energy subspace [22] [25]. The full dynamics of this open quantum system are governed by the Lindblad master

equation, which describes the non-unitary evolution of the system's density matrix, ρ, accounting for irreversible processes like decoherence and relaxation [1] [3].

A primary challenge in simulating such systems is the exponential growth of the Hilbert space with the number of qubits. For a 50-qubit system, direct diagonalization or exact propagation is computationally intractable. The MPO-TDVP framework provides a powerful approximation to overcome this curse of dimensionality. The core innovation of the proposed strategy lies in the efficient representation of the Lindblad dissipators within the MPO formalism [25] [40]. Each term in the Lindblad sum, which typically takes the form $\mathcal{D}[L_k]=L_k\rho L_k^\dagger-\frac{1}{2}L_k^\dagger L_k,\rho$, where L_k are the Lindblad operators, can be mapped onto a sparse MPO tensor acting on the doubled Hilbert space of the density matrix [39]. For a one-dimensional chain of qubits, this allows the entire Liouvillian superoperator, which generates the time evolution, to be represented as an MPO. This representation is particularly efficient for Gaussian environments like the Ohmic bath, where analytical progress can be made to construct highly accurate MPOs, sometimes leveraging infinite MPO (iMPO) evolution methods to approximate the semi-infinite bath structure [75]. This technique effectively casts the entire problem—system, environment, and their interaction—into a unified mathematical structure that can be handled numerically.

The time evolution itself is carried out using the 2-site Time-Dependent Variational Principle (TDVP). TDVP is a variational integrator specifically designed for evolving Matrix Product States (MPS) [67] [73]. Instead of projecting the global equation of motion onto the tangent space of the MPS manifold, which can lead to instabilities, TDVP projects the exact Heisenberg equation of motion onto the local tangent spaces of each site or pair of sites [6]. The 2-site TDVP update focuses on two neighboring sites at a time, solving for the optimal local Hamiltonian generated by effective operators acting from the left and right on the pair, and then updating the tensors for those two sites accordingly [9]. This process is iterated by sweeping sequentially down the chain of 50 qubits [65]. This local, iterative nature makes TDVP computationally more efficient and numerically more stable than global methods like Krylov subspace propagation, especially for long-time simulations of one-dimensional systems [73]. Furthermore, the use of 2-site updates is crucial for capturing the complex correlation structures that develop in the system over time, which is essential for correctly calculating non-local observables like the branch-correlation witness `W` [65]. The initial state of the 50-qubit array, taken as the ground state of the uncoupled transmons, can be represented as a simple product-state MPS with a bond dimension of $\chi=1$, providing a clean starting point for the simulation [74].

The table below summarizes the key specifications of the proposed simulation framework, detailing the rationale behind each choice.

Parameter	Specification / Value	Justification
System Size	N=50 transmon qubits	Represents the proposed experiment; large enough for collective effects but small enough for MPO simulation [2].
Algorithm	2-site Time-Dependent Variational Principle (TDVP)	Efficient, stable, and preserves correlations well for one-dimensional open quantum systems [65] [73].
Bond Dimension	Truncated at $\chi \leq 256$	Critical for computational feasibility; must be validated against convergence of physical observables [74].
Environment	Ohmic spectral density with ω_c=5 GHz	Standard model for describing the low-frequency noise affecting superconducting qubits [7] [8].
Temperature	T = 15 mK	Cryogenic temperature to minimize thermal excitations and ensure ground state dominance [22] [25].
Dissipation	Represented via Lindblad master equation	Standard framework for open quantum systems; terms implemented as local MPO tensors for efficiency [25] [40].
Key Observable	Branch-Correlation Witness (W)	Must be preserved to validate physical fidelity; used as a convergence check for bond dimension [65].

In summary, the proposed MPO-TDVP framework represents the state-of-the-art for simulating the desired open quantum system. It combines a robust time-evolution algorithm (TDVP) with an efficient representation of environmental coupling (MPO for Lindblad operators). The method is theoretically well-suited to handle the complexities of the 50-qubit transmon array interacting with an Ohmic bath at cryogenic temperatures. However, the ultimate success of this strategy is not guaranteed and is critically dependent on the interplay between the system's inherent physics, which generates entanglement, and the imposed computational constraints, most notably the maximum bond dimension. The next section will analyze this critical trade-off in detail.

The Bond Dimension Constraint: A Critical Analysis of Computational Scalability

The single most significant constraint governing the feasibility of the proposed MPO-TDVP simulation is the requirement to truncate the bond dimension at a fixed value of $\chi \leq 256$ [74]. This constraint is a pragmatic necessity driven by the finite computational resources available, as both memory requirements and computational cost scale polynomially with χ. While a bond dimension of 256 is manageable for many one-

dimensional quantum systems, it may be insufficient to accurately capture the full entanglement dynamics of a 50-qubit chain interacting with an environment, especially over long times. The adequacy of this bond dimension is the central unknown that determines whether the simulation can produce reliable results for the screening coefficient α_{screen}.

The fundamental challenge addressed by the MPS/TDVP framework is the efficient approximation of quantum states that possess limited entanglement. In many physical systems, particularly in one-dimensional gapped phases of matter, the entanglement entropy across a cut follows an area law, meaning it saturates to a constant value independent of the subsystem size. This property allows the MPS representation to capture the state's physics with a bond dimension that does not grow exponentially with system size. However, several factors relevant to the current problem can lead to a violation of the area law or generate a high degree of entanglement that requires a larger χ to resolve accurately. First, the system is an open quantum system coupled to a dissipative bath. The interaction with the bath continuously generates entanglement between the qubits and the environment degrees of freedom, causing the entanglement entropy of the qubit chain to increase over time [1] [3]. Second, the system operates far from equilibrium. The preparation of the initial state (ground state) and its subsequent evolution under the full Hamiltonian and drive pulses constitute a non-equilibrium quench, which is known to generate significant amounts of entanglement [10]. Third, the presence of disorder or random fluctuations, such as low-frequency flux noise which is a major source of decoherence in transmon qubits [32], can drive the system into a Many-Body Localized (MBL) phase. In MBL systems, entanglement grows logarithmically with time, leading to a persistent source of complexity that can require very large bond dimensions to simulate accurately [10] [46].

Recent advancements in tensor network methods have demonstrated the power of these techniques for simulating various quantum systems, including spin-boson models and other open systems [40] [41]. However, even with these advances, scaling remains a formidable obstacle. One study on simulating coherent oscillations in analog quantum simulators highlights the difficulty of maintaining coherence in large systems, pointing to the subtlety of decoherence mechanisms in superconducting qubits [46] [68] [69]. Another report on simulating a system of six cylinders notes that qualitative convergence could only be achieved when the required MPS bond dimension would exceed available computational resources, explicitly demonstrating the scaling challenges of the method [66]. This underscores the fact that while the MPO-TDVP algorithm is powerful, its practical application is always bounded by the amount of entanglement the chosen bond dimension can support.

The literature provides some guidance on what bond dimensions are typically required for similar problems. Simulations of finite-temperature dynamics in spin chains using TD-DMRG (a precursor to TDVP) often require bond dimensions in the range of hundreds to thousands depending on the temperature and system size [6]. Simulations of transport properties in correlated electron systems also routinely use χ values up to several hundred [2]. Given that the current problem involves a larger system (50 qubits vs. typical spin chains) and includes the added complexity of a structured, non-Markovian bath, it is plausible that a bond dimension significantly larger than 256 would be necessary to achieve convergence for all physical observables. The risk is that the truncation to $\chi=256$ introduces systematic errors that corrupt the calculation of α_{screen} and the branch-correlation witness `W`. If the true bond dimension required for convergence is, for example, 1024, then the result obtained at $\chi=256$ is likely to be physically meaningless.

This constraint directly impacts the project's timeline and budget. Committing to cryostat fabrication without first establishing that the simulation results are converged with respect to χ would be a premature and high-risk decision. The outcome of the simulation is contingent on this parameter, and without a clear understanding of its scaling, the validity of the screening coefficient prediction remains uncertain. Therefore, the bond dimension constraint transforms the research goal from a simple simulation into a rigorous validation exercise. The success of the project at this stage depends entirely on whether the physical properties of the 50-qubit transmon array, when simulated under the specified conditions, can be captured within the limited entanglement space afforded by $\chi \leq 256$. This necessitates a detailed investigation into the growth of entanglement in the system and a systematic exploration of the simulation's dependence on the bond dimension, as outlined in the subsequent sections.

Validation Protocol: Ensuring Physical Fidelity through Convergence Criteria

Given the critical uncertainty introduced by the bond dimension constraint ($\chi \leq 256$), establishing a robust validation protocol is paramount to ensure the physical fidelity of the simulation results. The protocol must provide clear, quantitative criteria for determining when the simulation has reached a sufficiently accurate representation of the system's dynamics. The proposed validation strategy rests on two pillars: monitoring the "Entanglement Floor" (ϵ) and enforcing the preservation of the branch-correlation

witness (W). These two metrics serve different but complementary roles in assessing the simulation's accuracy and reliability.

The first pillar is the convergence criterion for the Entanglement Floor, defined as \epsilon < 10^{-12}. In an MPS simulation, the bond dimension truncation occurs after each time step via a Singular Value Decomposition (SVD), where the smallest singular values of the bond matrix are discarded [76]. The sum of the squares of these discarded singular values represents the "entanglement production" at that step—that is, the amount of entanglement that is lost due to the truncation. The Entanglement Floor is the regime where this production rate becomes negligible and remains so for a sustained period during the simulation [76]. By defining a strict tolerance for the entanglement production rate, we establish a quantitative measure of convergence. An entanglement floor criterion of \epsilon < 10^{-12} is a very stringent target, indicating that the simulation has captured the vast majority of the dynamically generated entanglement, and that any further increase in bond dimension would yield diminishing returns in accuracy [76]. This metric is purely computational; it ensures that the simulation is not being polluted by unphysical accumulation of truncated entanglement. If the simulation reaches a point where the entanglement production rate drops below 10^{-12} and stays there, it provides strong evidence that the bond dimension of 256 is sufficient to maintain a stable and accurate representation of the state's Schmidt rank throughout the time evolution. However, this criterion alone is not sufficient to guarantee physical correctness. It is possible to have a stable simulation with low entanglement production but still wrong physics if the underlying bond dimension was too small to capture the correct correlation structure from the outset.

This is where the second pillar of the validation protocol becomes indispensable: the preservation of the branch-correlation witness (W). Unlike the entanglement floor, W is a physical observable tied directly to the scientific goal of the experiment. Its exact definition is not provided in the context, but it is implied to be a measure of long-range correlations or a signature of the specific quantum phase relevant to gravity mediation [65]. The stability of W during the simulation acts as a powerful physical consistency check. The 2-site TDVP algorithm is known for its ability to conserve energy and preserve certain topological and correlation properties of the state more faithfully than some other time-evolution schemes [73]. As the simulation evolves, W must be calculated at regular intervals. If W remains stable and converges to a value consistent with physical expectations, it provides compelling evidence that the simulation is not only numerically stable but also physically meaningful. Conversely, if W exhibits unphysical oscillations, drifts, or fails to converge, it serves as a definitive signal that the bond dimension χ=256 is too small to capture the delicate correlation structures that give rise to the witness [65].

This creates a crucial feedback loop: the simulation cannot be considered converged until `W` stabilizes, and the stabilization of `W` is the ultimate proof of the result's validity.

The combined use of these two criteria forms a comprehensive validation protocol. The entanglement floor criterion (\epsilon < 10^{-12}) ensures numerical robustness, while the preservation of `W` ensures physical fidelity. Both must be satisfied for the simulation to be deemed successful. This dual-criteria approach mitigates the primary risk associated with the bond dimension constraint. It moves beyond simply asking "Is the simulation stable?" to ask the more profound question, "Is the stable simulation correct?". By tying the convergence of the simulation to the behavior of a key physical observable, the protocol provides a much stronger basis for trusting the computed value of the screening coefficient, α_{screen}. Without this physical consistency check, a result obtained at $\chi=256$ would remain suspect, and proceeding to hardware fabrication based on it would be unjustified. The following table outlines the role of each criterion in the validation process.

Validation Criterion	Type	Role in Protocol	Interpretation of Success
Entanglement Floor (ϵ)	Numerical	Ensures the simulation is not artificially heating due to truncation error.	The rate of discarded entanglement falls below 10^{-12} and remains there, indicating saturation of the bond dimension. [76]
Branch-Correlation Witness (W)	Physical	Ensures the simulation captures the correct long-range correlations and physical phase.	The value of `W` remains stable and converges to a physically meaningful value throughout the simulation. [65]

Ultimately, the success of the entire computational effort hinges on passing this validation protocol. Achieving a low entanglement floor without preserving `W` is akin to building a beautiful sandcastle that collapses the moment it rains. Conversely, having a physically correct `W` but a runaway entanglement floor indicates a fundamentally flawed numerical procedure. Only when both conditions are met can the project confidently assert that the simulation results are trustworthy and that the outcome of OP3 Stage 3 is a valid indicator of the experimental setup's viability.

Actionable Research Plan: A Campaign to Resolve the OP3 Gate Condition

To definitively determine whether the screening coefficient $\alpha_{screen}(50,T_\phi)$ meets or exceeds the critical threshold of 10^{-2}, a structured and methodical computational campaign is required. This campaign must be designed to rigorously test the simulation's

dependence on the bond dimension χ and validate its results against the physical fidelity criterion of the branch-correlation witness `W`. Proceeding to fabricate a cryostat without obtaining a conclusive answer from this campaign would be a premature and high-risk action. The immediate and necessary path forward is to execute this numerical validation protocol, which serves as a minimal, cost-effective prerequisite gate.

The recommended research plan consists of three sequential steps:

Step 1: Baseline Simulation at χ=256 The first step is to perform the full MPO-TDVP simulation using the specified parameters: a 50-qubit transmon array, an Ohmic bath with ω_c=5 GHz, at T=15 mK, and with the bond dimension truncated at its maximum allowed value, χ=256. This baseline simulation will provide an initial estimate for the screening coefficient, $\alpha_{screen}^{(256)}$. During this run, the entanglement production rate (ϵ) will be monitored to assess its behavior, and the branch-correlation witness, `W`, will be tracked over time. The primary purpose of this step is not to obtain a final, trusted answer, but to establish a reference point. It will reveal whether the simulation runs to completion without numerical instabilities and provide an initial value for α_{screen} and `W`.

Step 2: Convergence Testing with Increasing Bond Dimension The core of the campaign is to systematically investigate how the results change as the bond dimension is increased. This is done by re-running the simulation multiple times with progressively larger bond dimensions, such as χ=512, χ=1024, and potentially higher, if computationally feasible. At each step, the same observables—the screening coefficient $\alpha_{screen}^{(\chi)}$ and the branch-correlation witness $W^{(\chi)}$—are calculated. The results from these runs are then plotted as functions of χ. The goal is to observe convergence. If both $\alpha_{screen}^{(\chi)}$ and $W^{(\chi)}$ stabilize and vary by less than a predefined acceptable tolerance (e.g., 1%) over a range of increasing χ, this provides strong evidence that the simulation has converged. The converged value of α_{screen} would then be considered the reliable prediction from the simulation. This step directly addresses the central uncertainty of whether χ=256 is sufficient.

Step 3: Threshold Assessment and Reporting Based on the convergence plot from Step 2, a definitive conclusion regarding the OP3 Stage 3 condition can be drawn.

- **If the converged value of α_{screen} is $\geq 10^{-2}$:** The simulation successfully validates the project's premise. The numerical results provide confidence that the proposed experimental setup is capable of mediating a screened gravitational

interaction at the required strength. This outcome justifies proceeding to the next stage of development, which includes planning for hardware fabrication.

- **If the converged value of α_{screen} is $<10^{-2}$:** This represents a significant failure of the hypothesis. The numerical simulation indicates that the 50-qubit transmon array, under the specified conditions, does not generate a strong enough signal. In this case, committing to hardware fabrication would be ill-advised. The project would need to be re-evaluated, potentially involving theoretical re-modeling or changes to the device architecture before any further investment is made.
- **If the simulation fails to converge even with a bond dimension significantly larger than 256 (e.g., $\chi>2048$):** This outcome suggests that the problem is likely beyond the scope of current MPO-based methods for this system size. The entanglement generated in the system may be too complex to be captured efficiently. This would represent a fundamental limitation of the chosen simulation tool and would require a strategic reassessment of the project's computational approach.

This three-step campaign establishes a clear, auditable, and scientifically rigorous path to resolving the OP3 gate condition. It is designed to be a minimal, cost-effective simulation campaign that can definitively confirm or falsify the $\alpha_{screen}\geq10^{-2}$ threshold without unnecessary computational expenditure. The table below summarizes this actionable plan.

Step	Action	Purpose	Key Metrics to Monitor
1	Run baseline MPO-TDVP simulation at $\chi=256$.	Establish an initial reference point for α_{screen} and W. Ensure the simulation is numerically tractable.	$\alpha_{screen}^{(256)}$, $W^{(256)}$, Entanglement Production Rate (ϵ).
2	Perform a series of simulations with increasing bond dimension ($\chi=512,1024,\ldots$).	Systematically test for convergence of physical observables as a function of entanglement capacity.	$\alpha_{screen}^{(\chi)}$ vs. χ, $W^{(\chi)}$ vs. χ.
3	Analyze the convergence plot and make a final determination.	Decide whether the OP3 Stage 3 condition ($\alpha_{screen}\geq10^{-2}$) is met, failed, or is indeterminate.	Final converged value of α_{screen}, stabilization of W.

By following this structured plan, the project can transition from a state of uncertainty to a position of confident knowledge regarding the feasibility of its core premise. This numerical de-risking is the essential prerequisite to justifying the substantial financial commitment required for cryostat fabrication.

Risk Mitigation and Strategic Alternatives

While the proposed MPO-TDVP simulation campaign is the most direct path to resolving the OP3 gate condition, it carries inherent risks, primarily related to the unknown scaling of entanglement in the 50-qubit system. Should the simulation prove intractable or fail to meet the $\alpha_{screen} \geq 10^{-2}$ threshold, a set of risk mitigation strategies and alternative approaches must be considered to safeguard the project's objectives. These alternatives aim to either salvage the core physics by adjusting the simulation parameters or explore different avenues for validating the experimental premise.

One of the most significant risks is that the converged value of α_{screen} is found to be below the 10^{-2} threshold. The white paper suggests that utilizing a 50-qubit spin-squeezed state can improve the required integration time by a factor of approximately 50x, reducing the detection window from 17 seconds to ~0.34 seconds [4]. This implies that the screening coefficient is directly proportional to the effective signal strength. If the simulation yields a value of $\alpha_{screen} = 10^{-3}$ instead of 10^{-2}, the signal would be an order of magnitude smaller. This would dramatically increase the required integration time or place a much higher demand on the effectiveness of the spin-squeezing enhancement, potentially making the experiment infeasible. To quantify this risk, a sensitivity analysis should be performed as part of the main simulation campaign. This analysis would calculate the required integration time for a range of α_{screen} values below the threshold, providing a clear picture of the project's vulnerability.

Another major risk is that the simulation fails to converge even with a bond dimension significantly larger than 256. This would indicate that the problem is computationally intractable with the current MPO-based tools for the specified system size. In this scenario, the project team must pivot to alternative strategies. Several options exist:

1. **Reducing System Size for Extrapolation:** A viable alternative would be to reduce the number of qubits in the simulation, for example, to $N=30$ or $N=40$. For a smaller system, a bond dimension of $\chi=256$ is more likely to be sufficient to capture the relevant physics. The simulation can be run to convergence for this smaller array. The screening coefficient, $\alpha_{screen}(N)$, can then be studied as a function of system size N. By fitting this data to a plausible functional form, one could extrapolate the expected value for the target system size of $N=50$. While this approach introduces uncertainties related to the extrapolation, it provides a way to gain valuable insight into the system's behavior without being blocked by the computational bottleneck.

2. **Exploring Alternative Algorithms:** The field of tensor networks is continually evolving, and other algorithms might offer better performance for this specific problem. For instance, Time-Evolving Block Decimation (TEBD) is another common algorithm for MPS time evolution, though studies suggest TDVP often performs better for Schrödinger picture evolution [73]. Other advanced tensor network ansatzes, such as Projected Entangled Pair States (PEPS) for higher-dimensional lattices or multi-scale entanglement renormalization ansatz (MERA), could be explored, although they come with their own significant computational overhead. A preliminary investigation into these alternatives should be undertaken if the standard MPO-TDVP approach proves unsuccessful.

3. **Developing Simplified Analytical Models:** If numerical simulations at the full system size are not feasible, developing simplified analytical or semi-analytical models can provide valuable insights. These models might involve mean-field approximations, perturbation theory around a solvable limit, or mapping the problem to a more tractable model like the spin-boson model [40] [41]. Such models would not replace the need for a full simulation but could help to understand the expected scaling of α_{screen} with system parameters (coupling strength, bath spectral density, etc.) and guide future numerical efforts.

4. **Hardware-In-the-Loop Prototyping:** In parallel with the main simulation campaign, a small-scale experimental prototype could be developed. This would involve fabricating and testing a chip with a few transmon qubits coupled to a simplified version of the shielding and readout apparatus. Measuring the relevant observables on this smaller system could provide empirical data to benchmark the simulation codes and build confidence in their predictive power.

The following table outlines the potential risks and corresponding mitigation strategies.

Risk	Probability	Impact	Mitigation Strategy
Failure to Meet α_{screen} Threshold	Medium	High	Perform sensitivity analysis on required integration time; Re-evaluate theoretical model or device design.
Insufficient Bond Dimension (\chi > 256)	High	Critical	Reduce system size (N<50) and extrapolate results; Explore alternative tensor network algorithms.
Simulation Intractability (χ unconverged)	Medium	Critical	Develop simplified analytical models; Initiate small-scale experimental prototyping.
Unmodeled Systematic Errors	High	Medium	Integrate residual force gradient from shielding analysis into the Bayesian statistical model .

These risk mitigation and alternative strategies provide a safety net for the project. They ensure that even if the primary path encounters obstacles, there are logical and scientifically sound pathways forward. The key is to begin this preparatory work in parallel with the main simulation campaign, allowing for a rapid pivot if needed. This proactive approach to risk management is essential for navigating the challenges of such a complex and ambitious project.

Final Synthesis and Go/No-Go Recommendation

The objective of this research was to determine the feasibility of achieving a screening coefficient $\alpha_{screen}(50, T_\phi) \geq 10^{-2}$ using the specified MPO-TDVP simulation and to establish whether this threshold justifies proceeding to cryostat fabrication. The analysis confirms that the proposed computational strategy is theoretically sound and represents the state-of-the-art for simulating the targeted open quantum system. The MPO-TDVP framework, with its 2-site updates and MPO representation of Lindblad dissipators, is an appropriate and powerful tool for modeling the 50-qubit transmon array coupled to an Ohmic bath at 15 mK [40] [73]. However, the success of this strategy is not guaranteed and is critically dependent on the system's intrinsic entanglement generation and its compatibility with the imposed bond dimension constraint of $\chi \leq 256$.

The core of the uncertainty lies in the scaling of entanglement. While a bond dimension of 256 is computationally manageable, it may be insufficient to accurately capture the complex, time-dependent correlations that arise from the non-equilibrium dynamics and dissipative coupling in the 50-qubit system [10] [66]. The literature provides numerous examples where MPS methods face severe scaling challenges, highlighting that the assumption of sufficiency for $\chi = 256$ is precisely that—an assumption, not a proven fact [66]. Therefore, the project cannot proceed based on a single simulation run at this bond dimension. The validity of any result is contingent upon a rigorous validation protocol.

The proposed validation protocol, which combines the monitoring of the entanglement floor (\epsilon < 10^{-12}) with the enforcement of the physical consistency criterion of the branch-correlation witness `W`, provides a robust mechanism for ensuring the simulation's fidelity [65] [76]. The stabilization of `W` serves as the ultimate arbiter of physical correctness, signaling that the simulation has captured the essential correlation structure of the system. Consequently, the outcome of the entire computational effort hinges on the results of a multi-step convergence campaign, systematically varying the bond dimension χ until both numerical stability and physical consistency are achieved.

Based on this comprehensive analysis, the recommendation is clear and unequivocal:

Committing to cryostat fabrication at this juncture is premature and constitutes an unjustified risk. The project's core premise is contingent on the outcome of the OP3 Stage 3 simulation. Spending real resources on hardware before this critical numerical validation is complete would be a significant waste of funds if the simulation were to fail to meet the $\alpha_{screen} \geq 10^{-2}$ threshold.

The correct and necessary course of action is to **proceed immediately with the prescribed numerical campaign**. This involves executing the three-step plan: 1. Running a baseline simulation at $\chi = 256$. 2. Conducting a series of simulations with increasing bond dimensions to test for convergence of α_{screen} and `W`. 3. Using the convergence analysis to definitively confirm or falsify the screening coefficient threshold.

Only upon the successful completion of this computational gate—where the converged value of α_{screen} is shown to be $\geq 10^{-2}$—should the project authorize the allocation of resources for cryostat fabrication. This cautious, methodical approach ensures that the project proceeds on a foundation of validated scientific and computational evidence, thereby maximizing its chances of success and minimizing the risk of a costly failure.

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